

Global Facilities - Design & Construction Standard - Mechanical - Chilled Water and Condenser Water

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1.0 Purpose

- 1.1 This document serves as a System Standard document that provides specifications that shall be followed during the Design, Engineering, and Construction of new Chilled Water and Condenser Water Systems as part of new Greenfield FAB Construction projects.
- 1.2 The standard is applicable to Chilled Water (CHW) and Condenser Water (CW) systems.
- 1.3 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects.

2.0 Scope

- 2.1 This document covers the new Basebuild System installations only, as part of new Greenfield FAB Construction projects. This document does **NOT** cover tool installation.
- 2.2 This document does **NOT** cover Glycol Chilled Water systems.
- 2.3 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.

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3.0 Definitions and Acronyms

Term/Acronym	Definition / Description
CHW	Chilled Water
CW	Condenser Water
FAT	Factory Acceptance Test
FMCS/SCADA	Facility Monitoring and Control System / Supervisory Control and Data Acquisition
IST	Integrated System Test
ITP	Inspection and Test Plan
LOTO/COHE	Lockout Tagout / Control of Hazardous Energy
LTCHW	Low Temperature Chilled Water
MTCHW	Medium Temperature Chilled Water
N+1+F	System sizing concept for equipment quantity consisting of N duty units, + 1 redundant standby unit, + future unit(s)
OAT	Operational Acceptance Test
PLC	Programmable Logic Controller
POC/LPOC	Point of Connection / Lateral Point of Connection
SAT	Site Acceptance Test
SCHW	Secondary Chilled Water
TUM	Tool Utility Matrix
UPS	Uninterruptible Power Supply
VSD/VFD	Variable Speed Drive / Variable Frequency Drive
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team

4.0 Roles and Responsibilities

Roles	Responsibilities
GFTT Lead (Discipline Engineer)	- Develop and maintain System Standard document - Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard - Perform periodic document review and update System Standard
FCT Lead (Discipline Engineer)	- Review and provide technical feedback on System Standard - Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	- Review and provide technical feedback on System Standard - Incorporate System Standard into Project Engineering, Design, and Construction - Document project deviations, exceptions, and exclusions from System Standard - Share new lessons learned to GFTT Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	Use System Standard as technical reference

5.0 Overview of Systems

5.1 The Chilled Water system is used to produce chilled water as the cooling medium for the FAB Cleanroom, Facilities systems, and general building spaces. The Condenser Water system is used to transfer the heat from the chillers to the cooling towers, and then to reject the heat to the outside atmosphere. Condenser Water is also used for process cooling of some Facilities equipment.

5.2 The main CHW & CW System equipment shall be as follows:

5.2.1 Chillers

5.2.2 Cooling Towers

5.2.3 Distribution Pumps

5.3 The auxiliary CHW & CW System equipment shall be as follows:

5.3.1 Heat Exchangers

5.3.2 Expansion Tanks

5.3.3 Condenser Water Sidestream Filtration System

5.3.4 Water Quality and Chemical Dosing Skids

6.0 Performance Specs

- 6.1 The Chilled Water Systems shall be designed to meet the following targets: See **Tables 1 to 3**.

Table 1: Design Performance Parameters for LTCHW

Parameter	Design Target	Remarks
Supply Temperature	5.6 – 6.0 °C (42.0 – 42.8 °F)	Recommended is 6.0 °C (42.8°F)
Supply Temperature Target Range	±1 °C (±1.8 °F)	Target Temperature Control Range during system operation
LTCHW ΔT Temperature	6.0– 8.0 °C (10.8 – 14.4 °F)	Recommended is 7.0 °C ΔT (12.6°F)

Table 2: Design Performance Parameters for MTCHW

Parameter	Design Target	Remarks
Supply Temperature	12 - 13 °C (53.6 - 55.4 °F)	Recommended is 13.0 °C (55.4°F)
Supply Temperature Target Range	±1 °C (±1.8 °F)	Target Temperature Control Range during system operation
MTCHW ΔT Temperature	6.0– 8.0 °C (10.8 - 14.4 °F)	Recommended is 7.0 °C ΔT (12.6°F)

Table 3: Design Performance Parameters for CW

Parameter	Design Target	Remarks
Supply Temperature	Depends on Ambient Design Wet Bulb (WB) Temperature	
ΔT Temperature	Depends on Ambient Design Wet Bulb (WB) Temperature	Recommended is 6.0 °C ΔT (10.8°F)
Cooling Tower Approach Temperature	2.5 - 3.5 °C (4.5 - 6.3 °F) To Ambient WB	Recommended is 3.0 °C ΔT (5.4°F) To Ambient WB Temperature

7.0 Equipment Sizing & Design Requirements

7.1 System Type for Temperature Supply – Chilled Water System shall be designed as two separate temperature Chilled Water loops; Medium Temperature Chilled Water (MTCHW) and Low Temperature Chilled Water (LTCHW).

7.1.1 The MTCHW chillers can operate ~25% more efficiently versus the LTCHW chillers, both at Full Load conditions and Part Load conditions.

7.1.2 For overall CHW Plant efficiency, a 2-temperature CHW system (MT/LT) can operate ~15-20% more efficiently versus a 1-temperature CHW system (LT only), with all other factors remaining equal.

- 7.2 **Load Distribution Split** – The load distribution between MTCHW and LTCHW should be approximately 70%MT:30%LT and 80%MT:20%LT.

In general, the following Chilled Water loads shall be distributed across each system as shown in **Table 4**.

7.2.1 Because the MTCHW is much more energy efficient than LTCHW, it is preferred to connect as much load as possible to MTCHW instead of LTCHW.

7.2.2 The LTCHW loads should be limited mostly to loads that require dehumidification or colder supply temperatures.

Table 4: Low Temperature versus Medium Temperature Loads

Low Temperature Chilled Water System	Medium Temperature Chilled Water System
- Cleanroom MAU Secondary Cooling Coils - Other Critical MAU or GHVAC Cooling Coils with de-humidification requirements	- Cleanroom MAU Primary Cooling Coils - Cleanroom Dry Cooling Coils - Non-Critical MAU or GHVAC Cooling Coils - Process Cooling Water - CDA & HPCDA Aftercoolers - Ultra Pure Water - Other Process Cooling

- 7.3 **System Type for Pumping Distribution** – Chilled Water System shall be designed as Variable Primary Type.

7.3.1 Variable Flow Primary & Primary/Secondary pumping configuration through the Chillers can operate ~3-5% more efficiently versus Fixed Flow Primary pumping configuration, for the entire CHW Plant Energy Consumption.

7.3.2 Secondary pump is only allowed for different consumer with higher pressure drop which required secondary pump to overcome it.

- 7.4 **Redundancy & Future** – CHW and CW System main equipment shall be designed for N+1+2F (for MTCHW) and N+1+1F (for LTCHW); for Chillers, Pumps, and Cooling Towers. All main equipment shall have N+1 redundancy (with one redundant unit having capability as online standby), and a provision to install one or two additional future units (1F or 2F) for future capacity expansion.

- 7.5 **Diversity and Safety Factor** – Chilled Water systems shall be sized so that the N capacity meets 115-120% of the Chilled Water demand at full build (after initial construction of the system). Therefore, the Chilled Water system sizing for N capacity shall have 15-20% spare future capacity over the demand at full build (after initial construction of the system). See **Table 5**.

Table 5: System Sizing Calculation

- MTCHW and LTCHW System N Capacity,
$$N = 115\text{-}120\% \times \text{Peak Summertime Cooling Demand}$$
- LTCHW System Sizing = $N + 1 + F$
- MTCHW System Sizing = $N + 1 + 2F$

- 7.6 **Equipment Quantity** – CHW and CW System main equipment shall be selected for a minimum of 3 units (2+1) or more, with each unit to be equal in capacity sizing; for the Chillers, Pumps, and Cooling Towers.
- 7.7 **Design Conditions for Equipment Sizing** – CHW System sizing shall be based on the total cooling load calculated during peak summertime conditions, as determined by the ASHRAE 0.4% Climatic Data for the site location.
- 7.8 **Primary Chilled Water Pump Arrangement** – There shall be one variable flow, Primary Chilled Water pump for each chiller. The design flowrate rating of the CHW Pump shall match the design evaporator flowrate for each chiller, with maximum capability to provide up to 120% of the design evaporator flowrate of each chiller. The Primary Chilled Water pumps shall be arranged in parallel together and connected via a common piping header.
- 7.8.1 There shall be one common piping header for the Primary LTCHW pumps to connect to the LT Chillers, so that any configuration of Primary LTCHW pumps can serve any LT Chillers.
- 7.8.2 There shall be one common piping header for the Primary MTCHW pumps to connect to the MT Chillers, so that any configuration of Primary MTCHW pumps can serve any MT Chillers.
- 7.8.3 By selecting the Primary Chilled Water pump with maximum 120% of the design evaporator flowrate of each chiller, the flowrate can be increased if needed to reduce the effects of chiller “Low-Delta T syndrome”.

7.9 Primary Condenser Water Pump Arrangement – There shall be one variable flow, Primary Condenser Water pump for each chiller. The design flowrate rating of the Primary CW Pump shall match the design condenser flowrate for each chiller. The Primary CW pumps shall be arranged in parallel together and connected via a common piping header, serving both the MT and LT Chillers.

7.9.1 There shall be one common piping header for all Primary Condenser Water pumps to connect to the MT Chiller and LT Chillers. Any combination of Primary CW pumps can serve any of the MT Chillers or LT Chillers.

7.10 Secondary Chilled Water Pump Arrangement– When there is a higher pressure drop demand compared to main demand, like a demand from different building. It recommends installing Secondary Chilled Water Pumps (SCHW), sized to match the flowrate demand and pressure drop of the connected loads for energy saving purpose. The SCHW Pumps shall be arranged in parallel together and connected via a common piping header.

7.11 Chiller Arrangement – The chillers shall be arranged in parallel together and connected via a common piping header, one header for MT Chillers and one header for LT Chillers.

7.11.1 There shall be one common piping header for the LT Chillers, so that any configuration of Primary LTCHW pumps can serve any LT Chillers.

7.11.2 Each LT Chiller shall have modulating type control valves on the LT Chilled Water supply piping and Condenser Water Supply piping, at the discharge side of each chiller. These control valves shall be programmed to open and enable flow upon chiller startup, and close to isolate flow during chiller shutdown.

7.11.3 There shall be one common piping header for the MT Chillers, so that any configuration of Primary MTCHW pumps can serve any MT Chillers.

7.11.4 Each MT Chiller shall have modulating type control valves on the MT Chilled Water supply piping and Condenser Water Supply piping, at the discharge side of each chiller. These control valves shall be programmed to open and enable flow upon chiller startup, and close to isolate flow during chiller shutdown.

7.12 Cooling Tower Arrangement – The cooling towers shall be arranged in parallel together and connected via a common piping header, serving both the MT and LT Chillers.

7.12.1 There shall be one common piping header for all Cooling Towers to connect to the MT Chiller and LT Chillers. Any combination of Cooling Towers can serve any of the MT Chillers or LT Chillers.

7.12.2 Each Cooling Tower shall have open/close type control valves on the Condenser Water Supply and Return piping of each cooling tower. These control valves shall be programmed to open and enable flow upon cooling tower operation, and close to isolate flow during cooling tower shutdown.

7.13 **Cooling Tower Equalizer Line** - An equalizer/balancing line shall be installed and connected to each of the sumps from the cooling towers. The equalizer line shall be sized so that it is at least as large as the largest line coming into the cooling towers.

7.14 **Equipment Layout** – Equipment shall be suitably located with sufficient space to comply with all equipment maintenance clearances and access requirements. Adequate space and clear routing path shall be reserved for future move-in of new equipment, to allow chiller tube removal, and to allow move-in and move-out of large components during maintenance and repair (such as motors, compressor, etc.).

7.14.1 Enough space is provided at each end of the chiller to allow removal of the evaporator and condenser tube bundles for cleaning and maintenance.

7.14.2 Dedicated hoisting beam(s) shall be provided to allow ease of removal and re-installation of large chiller components, such as the compressor and motor, during maintenance and repair.

7.15 **Pressure Drop Calculations** – Submit pressure drop calculations for each piping system, to show that each pump selection and the pump parameters (flow, pressure, power, NPSH, etc.) are adequate to meet the design requirements.

7.16 **Equipment Selection Criteria** – See **Table 6**.

Table 6: Equipment Selection Criteria

Equipment	Type
Chiller(s)	• Chillers shall be Centrifugal, water-cooled type. • Chillers shall consist of the following components: ○ Evaporator ○ Condenser ○ Compressor ○ Refrigerant ○ Microprocessor based control system with HMI Control Panel

Equipment	Type
	• It recommends design Across-the-Line, Full Voltage, no local starter type. Because medium-voltage starters by nature use less current than low voltage, and therefore have significantly less current inrush which will not be an issue for New fab design. • Chiller shall be specified with the following options/accessories: ○ Proof-of-flow proving switches hard wired to prevent the compressor from starting until chilled water flow has been established. ○ Differential pressure (flow proving) switches to ensure that minimum flowrate is met for chiller to operate ○ Factory-installed insulation on the evaporator section. ○ Anti-corrosion painting/coating on the chiller tube sheets and end-caps. ○ Self-closing pressure relief valves in lieu of burst discs. • Chiller technical submittal shall include the following information: ○ Full Load KW/Ton rating ○ NPLV/IPLV Part Load KW/Ton rating ○ Chilled Water entering and leaving water temperatures, flowrate, and pressure drop ○ Condenser Water entering and leaving water temperature, flowrate, and pressure drop ○ Evaporator and Condenser tube diameter, thickness, and fouling factors ○ KW/Ton Performance curve at different chiller loading and entering condenser water temperatures ○ Chiller minimum loading ○ Chiller sound level performance • Chiller installation shall be as follows: ○ Chillers shall be installed so that all required clearances and access for maintenance shall be free from obstructions to allow repair, maintenance, and component replacement. Enough space is provided at each end of the chiller to allow removal of the evaporator and condenser tube bundles. ○ Chilled water and condenser water piping shall be installed with full size isolation valves, pressure gauges, temperature gauges, and flexible connections. ○ Chillers shall be mounted on minimum 4" high concrete housekeeping pads with seismic restraint and vibration spring isolators. • The chiller motor shall be refrigerant cooled type or water-cooled type (using chilled water as the cooling source). Air-cooled type motors shall be not be used, due to the amount of heat discharged to the chiller room.

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Equipment	Type
	Note: Approved Manufacturers – JCI/York, Trane, Carrier, MHI, or approved equivalent
Distribution Pump(s)	• Chilled Water and Condenser Water Pumps shall be Centrifugal Type, End-Suction, or vertical inline or Split Case Type. • Pump sizing and selection shall be as follows: ○ Pumps shall be selected with the performance indicated at the design operating point (flow and pressure, at partial VSD speed and motor brake-horsepower). ○ Pump curve shall show the performance indicated at the maximum operating point (flow and pressure, based on pump impeller size, full VSD speed, and full brake-horsepower). ○ Pumps shall be Cast Iron casing, with Bronze, Carbon Steel, or Stainless Steel internally wetted parts (shaft, impeller, etc.). • Pump motor sizing and selection shall be as follows: ○ Pump motors shall be selected for non-overloading operation on all points of the performance curve at the design conditions. ○ Pump motors shall be rated for inverter duty with variable speed operation. ○ Pump motor efficiency rating shall be minimum IE3. ○ Pump motor IP rating shall be minimum IP55. • Pump installation shall be as follows: ○ Each pump shall be installed with full size isolation valves, pressure gauges, flexible connections, and low point drain valves on both the return and discharge side of the pump. ○ Each pump shall have a strainer or suction diffuser installed on the pump suction side, and a check valve installed on the pump discharge side. ○ Maintain a minimum of 5 pipe diameters of straight pipe immediately upstream of pump suction unless using a suction diffuser. ○ Each pump shall have concrete inertia base and vibration isolation springs. ○ Each pump shall be installed on minimum 4-inch (100mm) high concrete housekeeping pad. • Primary Chilled Water Pump selection shall include the following considerations:

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Equipment	Type
	○ Primary CHW Pump maximum flowrate shall be 120% of the chiller's design evaporator flowrate. (See Section 7.8) • Condenser Water Pump selection shall include the following considerations: ○ CW Pump pressure rating shall cover system supply pressure equal to the sum of the piping pressure loss, elevation head, condenser barrel pressure drop, and the cooling tower distribution header. ○ CW pumps cooling towers shall be located to ensure that condenser water pumps have sufficient net positive suction head (NPSH). The condenser water pumps should be located below the cooling tower sumps to ensure flooded pump suction lines. Note: Approved Manufacturers – Grundfos/Paco, Andritz, KSB, Armstrong, Bell & Gossett, Goulds, Peerless, Ebara, Hitachi
Cooling Tower(s)	• Cooling Towers shall be Crossflow or Counterflow type. • Cooling Tower shall consist of the following components: ○ Gravity water distribution system ○ Basin sweeper piping (with venturi type nozzles) filtration package ○ Float-operated mechanical make-up water fill valve assembly with bypass ○ PVC fill surface with integral drift eliminators ○ Air inlet louvers ○ Cold water basin with suction strainer and anti-vortex hood ○ Hot water basin ○ Sump bleed line and controls ○ Equalizer pipe (for multiple cooling towers piped together) • Cooling Tower shall consist of the following materials of construction: ○ Galvanized Steel or Fiberglass Reinforced Plastic (FRP) Frame ○ Stainless Steel or Concrete Lined Cold Basin ○ PVC fill, which shall be Factory Mutual (FM) approved ○ Non FM approved CT is allowed with CT sprinkler protection. ○ Default CT material will be Galvanized Steel, sprinkler are not required if the selected CT is FM approved CT. ○ Any alteration on the Galvanized Steel CT and void the FM approval certificate shall be compensate by the sprinkler protection.

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Equipment	Type
	○ For those part that prone to corrosion can be FRP coated, however this will void the FM approval and required sprinkler protection for CT. ○ Installation of CT Sprinkler protection shall follow recommendation of FMDS 01-06. • Cooling Tower sizing and selection shall be as follows: ○ Fan motor efficiency rating shall be minimum IE3 ○ Fan motor IP rating shall be minimum IP55. • Provide the Cooling Tower with the following options/accessories: ○ Fan vibration switch, that will enable an alarm in SCADA if the fan's vibration is out of specification. ○ Cooling tower axial fan blades shall be corrosion resistant (i.e. FRP). ○ PVC fill and drift eliminators shall be Factory Mutual (FM) approved materials or with installation of fire sprinkler. ○ Electric heaters for each sump/basin, if installed in areas with cold weather climates that experience freezing temperatures ○ Sprinkler fire protection for CT assembly and motor ○ Basin Sweeper Piping system to use pressurized water streams to spray and clean the basin sump. General sizing guidelines for Basin Sweeper Piping system shall be as follows: ▪ Water depth less than 3 feet - 1 GPM/square foot basin area ▪ Water depth greater than 3 feet - 1.5 GPM/square foot basin area • Cooling Tower shall be installed as follows: ○ Provide flexible connections, isolation valves, and gauges on both Condenser Water Return and Supply side of cooling tower ○ Provide automatic flow control valves on both Condenser Tower Water Return and Supply Sides of Cooling Tower ○ Provide Temperature Transmitter at Cooling Tower Water Supply side ○ Provide drain and overflow drain piping from Cooling Tower • Each cooling tower shall have separate sumps and isolation valves to allow individual maintenance and repairs to be performed on the cooling tower. • Cooling towers shall be located such that their plumes cannot enter HVAC fresh air intakes.

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Equipment	Type
	For Sites located in regions at risk of typhoons, Cooling Tower reinforcement for high winds and typhoons shall be designed and installed. Note: Approved Manufacturers – BAC, Evapco, SPX/Marley or approved equivalent
Sidestream Sand Filtration Skid(s) (for CW System)	- Provide Packaged Sidestream Sand Filtration skid(s) for the Condenser Water piping loop, rated for 3-5% recirculation flowrate of the Condenser Water system flowrate. - The Sand Filtration Supply piping shall be discharged directly from a header into the Cooling Tower Sumps, and the Return Piping shall be drawn directly from the Cooling Tower Sumps to the main Return header. - The Sand filtration skid shall be rated for minimum 90% efficient at 5µm particle size. - The Sand filtration skid shall be provided with local control panel with automatic switching capability and backwash/blowdown cycle, circulation pump, check valves, pressure gauges, flexible connections, and isolation valves. - Each filter shall be installed with air vent port, drain port, and differential pressure gauge.
Expansion Tank(s) (for CHW system)	- Provide Expansion Tank(s) in CHW return main piping prior to suction side of distribution pumps - Expansion Tank(s) shall be Diaphragm or Bladder Type. - Expansion Tank(s) shall be provided with drain port, air charging fitting, isolation valves, pressure gauge, and pressure relief valve.
Water Quality and Automated Chemical Dosing Skid(s)	- Provide Automated Chemical Dosing System(s) for Automated monitoring, control, and dosing of chemicals to the Condenser Water and Chilled Water Systems. The Systems shall consist of the following: - Day Tanks with Level sensors and Level switch - Local Control Panel - Dual Automated Metering Pumps (N+1) with check valves and isolation valves, for each chemical - Containment - Manual pour/fill station - PH sensor and controller - Conductivity sensor and controller - Corrosion Coupon Rack - The following Chemical Dosing shall be provided for the Condenser Water Chemical Dosing System: - Corrosion and Scale Inhibitor

Equipment	Type
	○ Sodium Hypochlorite (NaClO) ○ Sodium Hydroxide (NaOH) ○ Biocide • The following Chemical Dosing shall be provided for Chilled Water Chemical Dosing System: ○ Corrosion Inhibitor ○ Biocide

8.0 Piping Design and Sizing Requirements

8.1 **Pipe Sizing** – CHW and CW Main Pipe Sizing shall be selected based on the full build capacity of the system equipment, including the future equipment (N+1F for LTCHW, and N+2F for MTCHW). It is recommended for the Pipe Sizing to be designed with maximum velocities/pressure drop not to exceed the tables below: See **Tables 7**

Table 7 – Pipe Sizing Criteria for CHW and CW Piping

CHW Pipe Size	Maximum Velocity	Maximum pressure drop
< 6" (< DN150)		≤ 300 (Pa/m)
6" to 8" (DN150 to 200)	≤ 2.3 m/s	
10" to 20" (DN250 to DN500)	≤ 2.8 m/s	
22" to 30" (DN550 to DN750)	≤ 3.0 m/s	
32" or larger (DN800 or larger)	≤ 3.5 m/s	

8.1.1 If the pipe sizing is designed with velocities/pressure drop exceeding Tables 7, the piping system pressure drop calculations shall demonstrate that the pump selections have adequate performance rating to meet the design parameters and system requirements.

8.2 Materials of Construction

8.2.1 CHW distribution piping shall be A53 Carbon Steel or ASTM A53B SCH 40 galvanized steel.

8.2.2 All CW distribution piping shall be A53 Carbon Steel, Schedule 40 or ASTM A53B SCH 40 galvanized steel.

a) All CW piping installed and located at the cooling tower areas shall be hot dip galvanized.

8.2.3 CHW and CW piping shall be fully welded as much as possible, to minimize the number of joints and flanges.

8.2.4 Victaulic type pipe joints, fittings, or components shall **NOT** be used for CHW or CW systems and piping.

8.3 Spare Valves for Main Expansion

8.3.1 Full-size valves shall be installed at all end-of-line at the main plantroom headers for chillers and pumps, to allow for future expansion.

8.3.2 Full-size valves shall be installed at all end-of-line at the main distribution piping at each building, to allow for future expansion.

8.3.3 All end-of-line butterfly valves shall be true bi-directional end-of-line rated (capable of positive shutoff in both directions), and lug style.

8.4 Chilled Water Piping

8.4.1 Bypass Piping for Variable Primary type CHW Systems –

There shall be a bypass line with two-way modulating control valve to connect between the CHWS side to the CHWR side. The bypass line shall allow the minimum evaporator flowrates of the chillers to be met, upon startup and low-load conditions.

8.4.2 **Makeup Water Piping (Normal Fill)** – There shall be a Normal Makeup Water pipe, connected to the main Chilled Water Return piping upstream of the Expansion Tanks, to allow automatic filling. ICW Water or other source shall be used as the normal fill makeup water source. The Normal-Fill Makeup Water piping shall be minimum 1" size (D25) and shall be provided with isolation valves, dual parallel check valve/backflow preventer & pressure regulating control valve stations with isolation valves and bypass, pressure gauges, flowmeter, and pressure-relief valve with relief discharge piping.

8.4.3 **Emergency Quick-Fill Makeup-Water Piping** – There shall be an Emergency Quick-Fill Makeup Water pipe to connect to the main Chilled Water Return piping, to allow manual filling during emergency. ICW Water or other source shall be used as the emergency quick-fill makeup water source. The Emergency Quick-Fill Makeup Water piping shall be minimum 2 ½" size (DN65) and shall be provided with manual isolation valve, check valve/backflow preventer, and pressure gauges.

8.4.4 **End of Line Bypasses** – There shall be end-of-line bypass piping with manual valves at the end of all Chilled Water Supply and Return main lines, to eliminate dead-legs and ensure circulation can be maintained in the system.

8.5 **Condenser Water Piping**

8.5.1 **Condenser Water Automatic Makeup Piping via Cooling Tower Sump** – There shall be a make-up water line provided to each of the cooling tower sumps, and connected to a self-contained float valve. The float valve shall operate automatically to maintain a minimum sump level. The float, level assembly, and linkages shall be stainless steel construction.

8.5.2 **Condenser Water Manual-Fill Makeup Piping via Cooling Tower Sump** – There shall also be a separate bypass/manual-fill make-up water line to each of the cooling tower sumps.

8.5.3 **Condenser Water Main Drain Piping** – There shall be a main blowdown/drain pipe, with automatic control valve, isolation valves, flowmeter, and bypass.

8.5.4 **Cooling Tower Equalizer Piping** – There shall be an equalizer line to connect between each Cooling Tower sump. The equalizer line shall be minimum full-line size to match the size of the CWS/R piping to any of the Cooling Towers.

8.5.5 **Cooling Tower Sump Drain Piping** – Each cooling tower sump shall have a drain pipe with isolation valve and flexible connection, for manual draining of the sump.

8.5.6 **End of Line Bypasses** – There shall be end-of-line bypass piping with manual valves at the end of all Condenser Water Supply and Return main lines and branch lines, to eliminate dead-legs and ensure circulation can be maintained in the system.

8.6 **Control Valve Bypass Piping** - Bypass piping with isolation valves shall be installed across the Makeup and Blowdown/Drain Automatic Control Valves, to allow future isolation for repair or replacement of the valves.

8.7 **Refrigerant Relief Valve Piping** - Refrigerant relief piping shall be extended from each chiller refrigerant relief valve to an external outside area, for safe relief of the refrigerant in case of over pressure.

- 8.8 **Piping Routing** – Chilled Water and Condenser Water piping shall not be routed above FAB Cleanroom space, Electrical rooms, Server rooms, or other critical areas in which condensation or water leakage can cause damage to critical equipment with significant impact. This includes the main piping, as well as branch piping to loads and equipment such as Dry Cooling Coils, Air Handling Units, Fan Coil Units, etc.

9.0 Additional Redundancy Design

- 9.1 **MT/LT Switchover Chiller** – Consider providing one MT/LT Switchover Chiller shall be provided with the capability to operate as either an LT Chiller or MT Chiller. The piping shall be configured to connect this MT/LT Switchover Chiller to both LT and MT common piping headers, with isolation valves to allow operation on either system.
- 9.2 **MT/LT Backup Heat Exchanger** – A minimum one MT/LT Backup Heat Exchanger shall be provided in case of low loading in early stage or during winter season. The Heat Exchanger will be connected to the MTCHW loop on one side and the LTCHW loop on the other side. This allows the Heat Exchanger to be used to create MTCHW from the LTCHW chiller, to serve as additional backup redundancy to the MT Chillers. Also, during winter seasons with low loading, the Heat Exchanger can be used to provide additional load on the LT Chiller, so it's minimum load requirements can be met and it can continue stable operation.

10.0 Sustainability Design

- 10.1 **Chilled Water Optimization** – Controls logic for optimized system performance and energy efficiency shall be provided as part of the Chilled Water system. Refer to the Section 16.0 "Control Sequence Philosophy" for additional details.
- 10.2 **Heat Recovery Heat Exchanger** – Consider providing a Heat Recovery Heat Exchanger on the Condenser Water Return piping after the chillers, to recover the waste heat before the Condenser Water is routed to the Cooling Towers.
- 10.2.1 The Waste Heat Recovery can be used for the following users:
- CDA Compressor Cooling
 - Preheat coils in Makeup Air Units
 - Reheat coils in Makeup Air units
- 10.3 **Heat Recovery Chillers** – In areas with cold weather climate conditions in which heating provisions is needed, consider specifying Double Bundle Heat Recovery Chillers instead of Cooling Only Chillers. These chillers have the ability to recover waste heat from the Condenser bundle, and reject the waste heat to a heat recovery loop instead of (or simultaneously) to the Condenser Water loop. Typical hot water supply temperatures generated can be 35-38°C, option to 40-43°C and may be

able to replace the need for boilers and separate hot water system.

10.4 Waterside Free-Cooling – In areas with cold weather climate conditions, consider installing a Waterside Free-Cooling Heat Exchanger and dedicated Free-Cooling CHW and CW pumps. The Heat Exchanger will be connected to Condenser Water piping from one or more Cooling Towers, which shall have the ability to be isolated and de-coupled from the other Cooling Towers during winter seasons, via motorized valves. The de-coupled Cooling Towers will be operated at colder supply temperatures to provide Free-Cooling to the Chilled Water loop, via the Free-Cooling Heat Exchanger. The dedicated CHW and CW pumps will operate to ensure flow through the Free-Cooling Heat Exchanger and loop.

10.4.1 Engineering evaluation should be performed to determine the cost savings and return on investment (ROI) achievable via using Waterside Free-Cooling strategy, based on annual profile of Wet Bulb temperature, Cooling Tower approach performance, and number of available bin-hours for Free-Cooling operation.

10.5 Refrigerant Selection – The refrigerant specified for Chiller equipment shall be Non-ODP (Ozone Depletion) and Low-GWP (Global Warming Potential).

10.5.1 The following are some non-ODP and Low-GWP, $GWP < 15$, acceptable refrigerants for use, but not limited to the following:

- R-514A
- R-1233zd
- R-1234ze

10.5.2 Chlorofluorocarbon (CFC) and Hydrochlorofluorocarbon (HCFC) refrigerant types shall **NOT** be used on Chiller equipment.

- R-134a (HFC type) shall no longer be used and specified in new Chiller equipment purchases, as this refrigerant has high GWP levels ($GWP = 1,300$) and is currently being Phased Out from Industrial use and availability.

10.6 VFD Chillers – In areas with cold weather climate conditions, ex. Taiwan, Japan, US, etc., consider install VFD chiller in LT chilled water system, which is possible to use low condenser water temperatures during winter season for improved efficiency.

11.0 Areas with Cold Weather Climate Design

11.1 Electric heaters shall be provided for each cooling tower sump / basin, to prevent freezing.

11.2 It shall be ensured that there is always minimum flowrate through chilled water coils that may be exposed to freezing conditions (such as coils in Makeup Air Handling units). Provide the minimum number of chilled

water distribution pumps with Emergency power, to ensure that minimum flowrate can be sustained during a loss of normal power. See **Section 14.3**.

12.0 Chiller Room Design

12.1 Refrigerant monitoring and detection

12.1.1 The chiller room shall be provided with a standalone refrigerant gas monitoring and leak detection system. The gas detectors shall be suitably located around the chiller's location as per manufacturer's recommendation to effectively detect refrigerant leaks from any of the chillers.

12.1.2 The standalone refrigerant gas monitoring and detection system shall comply with all applicable local authorities' requirement and ASHRAE Safety Code Standard.

12.1.3 The refrigerant gas monitoring unit shall provide outputs for remote monitoring and connection back to FMCS/SCADA.

12.2 Emergency Purge Exhaust fan ventilation

12.2.1 An emergency purge exhaust fan and ductwork shall be provided for use upon detection of refrigerant leak.

12.2.2 In the event that a refrigerant leak is detected, the mechanical ventilation system shall activate and the exhaust fan shall automatically start, to purge the room of any refrigerant gas. Visual warning lights and audible alarm shall be provided to indicate location of leaks within the plant area.

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13.0 Accessories

13.1 The following accessories shall be installed: See **Table 8**.

Table 8: Accessories to be installed

Accessories	Type & Installation Location	Remarks
Insulation	- All CHW distribution piping, piping components, and distribution pumps shall be fully insulated for thermal protection and to prevent condensation. - For pumps requiring insulation, provide insulated box "sections" for the pumps, to make pump access easier for maintenance.	In accordance with Spec Section 15260 (or equivalent Spec)
Piping Supports	- Install piping hangers, supports, racks, and clamps at the appropriate spacing interval as required by the system and piping load and layout - Provide seismic restraints and supports as deemed required during design, based on site location and local jurisdiction requirements - Provide vibration isolation for piping supports as deemed required during design, based on the building vibration criteria	In accordance with Spec Section 15140 (or equivalent Spec)
Flexible Connections	- Install at all Pump connections on suction and discharge - Install at all Chiller piping connections - Install at all Cooling Tower piping connections - Install at all Cooling Tower Sidestream Filtration Package piping connections - Install at all Heat Exchanger connections	
Air Vents	- Install Auto Vents at all high point locations of CHW and CW piping. - Installation of auto air vents require an isolation valve upstream of the vent for service and manual operation. - Install a section of clear PVC piping on the downstream side of the vent, for visual inspection. - If the main condenser water header centerline is at the same elevation of the centerline of the condenser pump suction, automatic air vents shall be installed on the pump casing to remove entrained air.	
Drain Ports	- Install at all low point locations - Install at all pump suction piping	

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Accessories	Type & Installation Location	Remarks
	Install at all filter housings	
Valves	- CHW and CW valves DN65 (2-1/2 inches) and smaller shall be full port ball valves with locking handles. - Valves DN80 (3 inches) and larger shall be ductile iron body, Ethylene Propylene Diene M (EPDM) class rubber liner, stainless steel disc and stem, lug style butterfly valves with position locking handle. - Wafer type butterfly valves shall NOT be used in CHW and CW systems. - CW valves 2-1/2 inches and smaller that are used for drains and vents and are open to freezing temperatures shall have a hold drilled in ball to allow water to drain from inside the ball to prevent ball freezing. - Self-closing pressure relief valves to be provided for chiller in lieu of burst discs. - Instrumentation points of connection shall be provided with a means of isolation (ball valve, etc.) for maintenance and/or replacement. - The Control valves shall be installed in such a way that it is able to view the valve opening status and its local indicator when standing at ground level. It does not recommend installing balancing valves to consumer with modulating control valve. - All valves (except control valves) should be the full size of the pipe before reducing the size to make connections at each load terminal. - Provide chain wheel operators for all valves 8 inches or larger mounted greater than 8 feet 0 inches above finish floor level. - Point-of-connection (P.O.C.) valves shall not be installed from the bottom of branch pipes to prevent a dirt leg which can trap debris.	
Labeling and Identification	"MTCHWS" and "MTCHWR" nomenclature shall be used for Medium Temperature Chilled Water Supply and Return "LTCHWS" and "LTCHWR" nomenclature shall be used for Low Temperature Chilled Water Supply and Return	In accordance with Spec Section 15190 (or equivalent)

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Accessories	Type & Installation Location	Remarks
	“CWS” and “CWR” nomenclature shall be used for Condenser Water Supply and Return - Provide identification labels at minimum of every 6m distance and change of direction - Labels shall be Green Background with White Lettering	Spec)

14.0 Electrical Design

14.1 **Power Segregation** – Electrical power supply for Chillers, Pumps, and Cooling Towers shall be segregated among a minimum of 2 separate upstream power sources (including panel + transformer).

14.2 **Un-interrupted Power for Controls** – 100% of CHW system controls including PLCs, Local Control Panels (LCPs), and other auxiliary control power shall be connected to an Uninterrupted Power source (UPS) to allow continuous operation of the CHW system during momentary power interruption or voltage sags, spikes, and fluctuation events.

14.3 **Emergency Power / UPS Power** – Refrigerant monitoring and safety ventilation equipment located in mechanical spaces shall be provided with backup or emergency power.

14.3.1 **Controls** - 100% of CHW system controls including PLCs, Local Control Panels (LCPs), and other auxiliary control power shall be connected to an Uninterrupted Power source (such as UPS or DC Bank) to allow continuous operation of the CHW system during momentary power interruption or voltage sags, spikes, and fluctuation events.

14.3.2 **Chilled Water Distribution Pump (Primary CHW Systems)**
– For Primary only CHW systems, a minimum of (1) Primary CHW pump shall be provided with emergency power backup. Upon a loss of normal power, the pump shall operate and provide minimum chilled water flow through the distribution network.

14.3.3 In areas with cold weather climate, ensure that there is always minimum flowrate through chilled water coils that may be exposed to freezing conditions. This includes providing pumps with emergency power to ensure that minimum flowrate can be provided during a loss of normal power.

14.4 **Variable Frequency Drives** – All Primary CHW pumps, Secondary CHW pumps, Primary CW pumps, and Cooling Tower fans shall be VFD driven for variable speed operation. Variable Frequency Drives shall be selected to be one size larger than the power rating of the actual motor size.

14.4.1 Variable Frequency Drives shall be compliant with SEMI standard F47 for voltage sag immunity, to provide ride-through capability during momentary power interruption or voltage sag.

14.4.2 Variable Frequency Drives shall be configured to "Fail Last Speed" configuration upon any failure, including but not limited to, loss of power, loss of communications, etc.

14.5 **Motor Efficiency** – Distribution pump and cooling tower fan motors shall be National Electrical Manufacturers Association (NEMA) or IE3 above.

14.6 **Local Disconnect Switches** – Each pump or cooling tower fan shall have a lockable type, local disconnect switch installed adjacent to the equipment, to allow for manual isolation and lockout/tagout (LOTO/COHE) of the equipment.

14.7 **Electrical / control Panel** – No top cable entry for all power and control to avoid water ingress. All cable should be mounted with cable lug termination in vibration isolated area.

15.0 Instrumentation

15.1 The following Instrumentation shall be installed and monitored/controlled via FMCS: See **Table 9**.

Table 9 – Instrumentation Devices

Accessories	Type & Installation Location
Pressure Transmitters	- Pressure Indicating Transmitters (PIT) shall be installed at the following locations: - Secondary supply header at the farthest downstream point, or the most hydraulically remote point for Primary/ Secondary CHW systems. - Primary supply header at the farthest downstream point, or the most hydraulically remote point for Primary Variable Flow CHW systems. - Main condenser water supply and return header piping. - Primary/Secondary CHW system and Variable Primary Flow CHW system distribution pump VSDs shall be used to maintain differential pressure set point at some selected point in the system.
Temperature Transmitters	Temperature Indicating Transmitters (TIT), to be used for "real time" monitoring, shall be installed at the

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Accessories	Type & Installation Location
	following locations ○ Main Chilled Water supply piping ○ Main Chilled Water return piping ○ Main Condenser Water supply piping ○ Main Condenser Water return piping ○ Main CHW supply header end of line (primary/secondary) at the farthest downstream point, or the most hydraulically remote point. ○ Wet bulb temperature sensor for cooling tower
Flowmeters	• An isolable insertion type MAG Flowmeter shall be installed on the main supply header to be used for "real-time" load monitoring. • Non-intrusive type flowmeters shall be installed at the following locations: ○ Chilled Water supply piping to measure total system flow. ○ Condenser Water supply piping to measure total system flow. ○ Each main Chilled Water supply line to measure chilled water flowrate to different buildings. ○ Condenser water system make-up water line to measure total system make-up water flow. ○ Condenser water system blowdown drain line to measure total system blowdown water flow. ○ Chilled Water return line for each chiller ○ Condenser Water return line for each chiller. • Locate all flow measuring devices in accessible locations with straight sections of pipe upstream (10 pipe diameters) and downstream (5 pipe diameters) of the device, or as recommended by the manufacturer.
PH sensor	• A pH sensor or probe shall be installed in the condenser water piping loop at the Automated Water Quality Skid, for monitoring of pH levels and to initiate chemical dosing.
Conductivity Sensor	• A Conductivity sensor or probe shall be installed in the condenser water supply loop, at the Automated Water Quality Skid, for monitoring of Conductivity levels and to initiate conductivity based blowdown/drain.

15.2 Instrumentation points of connection shall be provided with a means of isolation (ball valve, etc.) for maintenance and/or replacement.

16.0 Control Sequence Philosophy

16.1 The overall philosophy for system control and operation shall be listed in this section. Refer to the specific Sequence of Operations for each project for specific details.

16.2 Provide real-time monitoring analysis of CHW Plant operational status and kW/Ton performance.

16.2.1 The FMCS system shall calculate the CHW Plant "real-time" load (kW/Ton) based on readings from the system flowmeters, supply and return header temperature differential, and equipment power consumption data.

16.3 Chilled Water System Optimization logic

16.3.1 The Condenser Water flowrate shall be reset to maintain a consistent ΔT across the chiller, to reduce the Primary CW pump flowrate and provide pump energy savings. This shall be achieved by modulating the Primary CHW Pump speeds via the VSDs, based on the actual load and measured ΔT across the chillers.

16.3.2 The Cooling Tower fan speed shall be modulated based on the Condenser Water supply temperature setpoint and ambient wet bulb temperature, to provide fan energy savings. This shall be achieved by modulating the Cooling Tower fan speeds via the VSDs, based on the actual load, ambient wet bulb temperature, and CWS temperature setpoint.

16.3.3 The Condenser Water supply temperature setpoint shall be reset based on the ambient wet bulb temperature, to achieve the coldest Condenser Water possible while maintaining stable operation of the Chillers. This will provide energy savings through more efficient chiller operation.

16.3.4 Maximize CHW plant efficiency by using predictive analysis models and system analysis for the following:

- Forecast plant loads based on weather conditions
- Choose the most efficient combination of equipment loading to match the plant load.
- Allow the chillers, cooling tower and pumps to operate at their most efficient range

16.4 Chiller Operation

16.4.1 Provide the controls interface and communication details to allow information to be communicated between the chiller control panel and FMCS/SCADA. Chiller operational information shall include, but is not limited to, vane opening percentage, compressor loading percentage, hot gas by-pass valve opening status, evaporating

pressure & temperature, condensing pressure & temperature, alarms and trip signals, etc.

- 16.4.2 The chiller's condenser approach temperature (temperature difference between the condenser's refrigerant temperature and the condenser water leaving temperature) shall be monitored to ensure that the condenser tubes are not becoming fouled.
- 16.4.3 The chiller evaporator and condenser water flowrates shall always be maintained between the chiller's minimum and maximum limits.
- 16.4.4 A minimum downtime permissive in the local chiller control panel shall prevent the chiller from restarting until it has been off for a minimum amount of time as determined by the operator.
- 16.4.5 Upon startup or shutdown of a chiller, the modulating control valves on the chiller's CHW supply piping shall be programmed to open partially in order to satisfy the minimum flowrate to allow startup of the chiller, but meanwhile limiting the impact of temperature fluctuation caused to the Chilled Water system. Upon startup, after the chiller has reached its supply temperature setpoint, then the modulating control valve can be opened up fully. Or upon shutdown, after the chiller's shutdown sequence is completed, then the modulating control valve can be closed fully.

16.5 Control Valves

16.5.1 The following control valve fail positions shall be configured: See **Table 10**.

Table 10: Control Valve Fail Positions

Control Valve	Control Valve Fail Position		
	Loss of Air	Loss of Electricity	Loss of Control Signal
LT/MT Chiller CHW and CW control valve on Supply piping from each chiller	Fail Last State	Fail Last State	Fail Last State
CHW Makeup Water Control Valve	Fail Closed	Fail Closed	Fail Closed
Bypass line control valve for modulating the minimum evaporator flowrate	Fail Last State	Fail Last State	Fail Last State
Quick-Fill Makeup Water Control Valve	Fail Closed	Fail Closed	Fail Closed
Cooling Tower Supply and Return control valves on piping to/from each CT	Fail Last State	Fail Last State	Fail Last State
Cooling Tower Makeup Water Control valve	Fail Last State	Fail Last State	Fail Last State
Conductivity Blow-Down Control Valve	Fail Closed	Fail Closed	Fail Closed

16.5.2 The control valves shall be double-acting type actuators, capable of failing to last state position and holding the position for unlimited duration without needing control air, power, or PLC signal.

16.5.3 After a loss event of control air, power, or PLC signal, and after it is restored, the control valve output command/position shall be set to the same command/position it was operating at prior to the loss event.

17.0 Inter-Discipline Coordination Matrix

17.1 The following provisions shall be provided by each of the following disciplines, and coordinated accordingly: See **Table 11**.

Table 11: Inter-Discipline Coordination Matrix

Discipline	Provision
Architectural	• Provide concrete housekeeping pad below all equipment including chillers, pumps, heat exchanger, filters, expansion tanks, etc. • It is recommended that each cooling tower have an independent basin which can be isolated and drained. The basin isolation wall shall be incorporated into cell separation walls (no carrier over from sumps). • Cut openings in walls and floors for piping penetrations, and seal openings after piping installation • Provide adequate fire stopping at all required fire wall penetrations • Provide Emergency Safety Showers nearby Chemical Dosing Skid(s) and at all other required locations
Mechanical	• Provide HVAC space conditioning for dedicated Utility Room or Mechanical Room space containing the CHW System equipment. • Provide emergency exhaust fans, ductwork, and grills for the Refrigerant monitoring and detection system • Provide floor drains located suitably to allow proper draining of equipment, piping, and air vents
Electrical	• Provide VSDs for all pumps and Cooling Tower Fans • Provide local disconnect switches for all pumps and Cooling Towers • Provide power wiring • Provide grounding • Provide Emergency Power / UPS Power as outlined in this Standard • Provide Power Segregation as outlined in this Standard
Water / Wastewater	• Provide drain piping to appropriate drain system for Cooling Towers and other equipment/piping as required. • Provide makeup water supply and drain piping for Cooling Towers. • Provide water supply and drain piping for Emergency Safety Showers.
Controls	• Provide control valve actuators and positioners • Provide control wiring • Provide instrument control air tubing • Provide FMCS/SCADA system monitoring and controls of all instrumentation and transmitters including pressure, temperature, flow, conductivity, etc.

18.0 Quality Inspection and Testing

18.1 **Inspection and Test Plan (ITP)** – Create formal Inspection and Test Plans (ITP) that outline all requirements for quality verification, inspections, testing, and commissioning:

18.1.1 Piping and Ductwork

- Piping and Ductwork Shop Drawings and Submittals
- Incoming Material Inspection
- Material Storage
- Welding Procedures, Qualifications, and Inspection
- Piping, Ductwork, and Supports Installation and Verification
- QA/QC Inspection & Punchlist
- Leak Testing and Pressure Testing
- Cleaning, Flushing, and Passivation
- Water Balance
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

18.1.2 Equipment

- Equipment Shop Drawings and Submittals
- Factory Acceptance Test
- Incoming Equipment Inspection
- Equipment Move-In and Installation
- QA/QC Inspection & Punchlist
- Equipment Pre-Startup Checks – Mechanical, Electrical, Controls, etc.
- Operational Acceptance Test (OAT) - Standalone Equipment Startup per Vendor's Procedure to allow equipment to run in MANUAL mode. Also considered "Early Run" Equipment.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

18.1.3 System

- Site Acceptance Test (SAT) – Test System operation and control functions in accordance with P&ID, SOO, and Specifications.
- Integrated System Test (IST) – Test full System operation including all system interdependencies, interlocks, and critical failure modes.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

18.2 **QA/QC Inspection & Punchlist** – Prior to filling, leak testing, and pressure testing, QA/QC inspection & punchlist shall be performed on all installed piping and components to verify proper installation in accordance with construction drawings, project specifications, and industry standards. Perform a final P&ID line check to ensure the installation has been properly reinstated after the pressure testing, cleaning, flushing, etc.

18.3 **Gross Leak Test** - Gross leak test shall be performed via compressed Nitrogen or Air at pressure up to 5 psig, on metallic piping only. Walk the entire system to verify that no leaks are present and make repairs to any leaks that are identified.

- Compressed Air or Nitrogen shall NOT be used on any plastic piping (i.e. PVC, CPVC, PP, PE, HDPE, ABS, and PVDF).

18.4 **Hydrostatic Pressure Test** - All piping shall be hydronic pressure tested at a minimum of 1.5 times the design operating pressure, for a minimum of 2-4 hours. Properly vent the system to remove air via all air vents, during the water filling process.

The actual hydronic pressure test procedure, including means, methods, testing pressures, and durations, shall be determined during the construction project based on project specific requirements. The procedure shall be reviewed and approved by Micron for use.

18.5 **Cleaning and Flushing** – Cleaning and flushing of the piping shall be performed to properly clean, passivate, and prepare the piping for operational use.

The actual cleaning and flushing procedure, including means, methods, and durations, shall be determined during the construction project based on project specific requirements. The procedure shall be reviewed and approved by Micron for use.

The following is one possible method of cleaning and flushing of the piping – See Section 18.5.1 to Section 18.5.8. Alternate methods may be proposed for each construction project.

18.5.1 Water Flushing – The initial flush is a mechanical flush to remove construction debris, dirt, mud, silts, coarse materials, and foreign particles.

- Fill the system using clean water without chemicals. Properly vent the system to remove air via all air vents, during filling process.
- Do **NOT** flush through heat exchangers and equipment.
- Install temporary provisions and spool pieces to bypass the equipment such as heat exchangers, chillers, and control valves.
- Circulate at a minimum velocity of 2.5 – 3.0 fps (0.6 – 0.9 m/s).

18.5.2 Drain the water used during initial Water Flushing.

18.5.3 Remove the temporary bypass provisions and re-install pipe connections to heat exchangers, chillers, and equipment.

18.5.4 Chemical Flushing – The second flush is a chemical flush to remove particulate iron, smaller particles, and iron deposits on the metal.

- Fill the system using water with corrosion inhibitor and dispersant. Properly vent the system to remove air via all air vents, during filling process.
- This chemical flush should be performed within 24 hours of the initial water flush.
- This chemical flush will flow through the heat exchangers and equipment.
- Circulate at a minimum velocity of 2.5 – 3.0 fps (0.6 – 0.9 m/s).
- Circulate for minimum 24 hours.
- Check and confirm iron content < 2 ppm

18.5.5 Drain the water used during second Chemical Flush

18.5.6 Chemical Pre-Treatment/Passivation – The final flush is a chemical pre-treatment/passivation circulation to passivate the metal and provide a protective oxide barrier to prevent corrosion.

- Fill the system using clean water with high dosage of corrosion inhibitor. Properly vent the system to remove air via all air vents, during filling process.
- This final flush will flow through the heat exchangers and equipment.
- Circulate at a minimum velocity of 2.5 – 3.0 fps (0.6 – 0.9 m/s).
- Circulate for minimum 24 hours.
- Check and confirm iron content < 2 ppm

18.5.7 Final Adjustment – After completion of the pre-treatment/passivation, the water can remain in the system. Adjust water quality to final levels, and provide final chemical dosing.

18.5.8 Final Approval– Obtain approval from the Water Treatment specialist and the Site Facilities Operations team on the water quality.

19.0 Startup Commissioning / Site Acceptance Test

19.1 At a minimum, the following system control functions shall be tested and successfully proven during the System Startup Testing & Commissioning / Acceptance Tests.

19.1.1 Pumps

- Pump Motor Rotation Check
- Pump/Motor Alignment Check and Lubrication
- VSD Settings and Functionality Checks
- Control interface and communication checks to FMCS/SCADA
- Pump Standalone Startup Procedure per Manufacturer's procedure
- Verify Pump Operation through its operating range (0 to 50/60 Hz)
- Verify Pump Rotation Start/Stop Function
- Verify Pump Auto-Start sequencing, upon failure of an operating Pump

- Confirm System pressure control is stable during pump manual start/stop rotation, and pump auto-start sequencing (upon failure of another pump)
- Verify Pump speed control in response to system pressure fluctuations

19.1.2 Cooling Towers

- Fan Motor Rotation Check
- Fan/Motor Alignment Check and Lubrication
- VSD Settings and Functionality Checks
- Control interface and communication checks to FMCS/SCADA
- Cooling Tower Standalone Startup Procedure per Manufacturer's procedure
- Verify Cooling Tower Fan Operation through its operating range (0 to 50/60 Hz)
- Verify Cooling Tower Rotation Start/Stop Function
- Verify Cooling Tower Auto-Start sequencing, upon failure of an operating Cooling Tower
- Verify Cooling Tower Fan speed control in response to temperature setpoint changes

19.1.3 Chillers

- Chiller Motor Rotation Check
- Control interface and communication checks to FMCS/SCADA
- Chiller Standalone Startup Procedure per Manufacturer's procedure
- Verify Chiller Rotation Start/Stop Function
- Verify Chiller Auto-Start sequencing, upon failure of an operating Chiller

19.1.4 Heat Exchangers

- **Verify Heat Exchanger Control Valve Function through the operating range**
- Confirm Heat Exchanger Control Valve control in response to system temperature fluctuations

19.1.6 Sidestream Sand Filtration Skid(s)

- Filter Skid Standalone Startup Procedure per Manufacturer's procedure
- Integrated System Operation

19.1.7 Water Quality and Chemical Dosing Skid(s)

- Chemical Dosing Skid Standalone Startup Procedure per Manufacturer's procedure
- Integrated System Operation

19.1.8 Instrumentation & Controls

- Confirm Auto-switch to backup pressure transmitter upon failure of main pressure transmitter
- Confirm Auto-switch to backup temperature transmitter upon failure of main temperature transmitter
- Verify all alarms are enabled and activated in SCADA

20.0 Reference Specifications

20.1 The following Construction Specifications shall be referenced during design, engineering, and construction. The Specifications shown below are in accordance with the Construction Specifications Institute (CSI) Master Format; Refer to the specific Construction Specifications for each project. See **Table 12**.

Table 12 – Specification Sections

Spec Section	Spec Contents	Remarks
Section 15864	Centrifugal Chillers	
Section 15713	Cooling Towers	
Section 15140	Piping Supports and Anchors	
Section 15190	Piping Labels and Identification	
Section 15260	Piping Insulation	
Section 15280	Equipment Insulation	
Section 15515	Hydronic Specialties	
Section 15990	Testing, Adjusting, and Balancing	
Section 15170	Motors	
Section 15171	Variable Frequency Drives	
Section 15060	Pipe and Pipe Fittings	
Section 15061	Carbon Steel Piping	
Section 15120	Piping Specialties	
Section 15101	Valves	
Section 15540	Pumps	
Section 11223	Filters	
Section 11075	Heat Exchangers	
Section 15242	Vibration Isolation	

21.0 Document Control

21.1 **Approval:** GLOBAL_FAC_GFTT_APPR

21.2 **Notification:**

GLOBAL_FAC_MANAGERS
 GLOBAL_FAC_COMMUNICATIONS
 GLOBAL_FAC_CONSTRUCTION
 FCG_F4_FAC_NOTIFY
 FCG_F6_FAC_NOTIFY
 FCG_F10_FAC_NOTIFY
 FCG_F11_FAC_NOTIFY
 FCG_F15_FAC_NOTIFY
 FCG_F16_FAC_NOTIFY
 FCG_MMP_FAC_NOTIFY
 FCG_MMY_FAC_NOTIFY
 FCG_MSB_FAC_NOTIFY
 FCG_MTB_FAC_NOTIFY
 FCG_MXA_FAC_NOTIFY

21.2 **Final Viewer Access**

All Micron Team Members at all Sites

21.3 **Retention**

Adherence to the requirements specified in

[Global Facilities - Document Control Procedures](#)

21.4 **Review**

Biennially (two years)

22.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0		New document	05/26/2020
1	6.0	Table 1: Design Performance Parameters for LTCHW: LTCHW ΔT Temperature:Design Target:change to "6.0– 8.0 °C(10.8 – 14.4 °F)" from "5.6 – 7.7 °C(10 – 14 °F)" LTCHW ΔT Temperature:Design Target:change to "6.0 °C ΔT(10.8°F)"from "7.0 °C ΔT(12.6°F)"	05/19/2022

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Revision #	Section Changed	Details of Changes	Date
	7.0	Table 2: Design Performance Parameters for MTCHW: MTCHW ΔT Temperature:Design Target:change to "6.0– 8.0 °C(10.8 – 14.4 °F)" from "5.6 – 7.7 °C(10 – 14 °F)" MTCHW ΔT Temperature:Design Target:change to "6.0 °C ΔT(10.8°F)"from "7.0 °C ΔT(12.6°F)" Table 3: Design Performance Parameters for CW Add ΔT Temperature requirement 7.3 System Type for Pumping Distribution: Remove "Chilled Water System shall be designed as either Variable Primary/Variable Secondary Flow type, or Variable Primary Type. The type of system shall be determined during project design, based on the actual project constraints including building space, system layout, piping routing, etc." Add 7.3.2 7.10 Secondary Chilled Water Pump Arrangement Change to "When there is a higher pressure drop demand compared to main demand, like a demand from different building. It recommends installing Secondary Chilled Water Pumps (SCHW), sized to match the flowrate demand and pressure drop of the connected loads for energy saving purpose. The SCHW Pumps shall be arranged in parallel together and connected via a common piping header"	

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		from "(for Primary/Secondary type CHW Systems) - Secondary Chilled Water Pumps (SCHW) shall be variable speed type, sized to match the flowrate demand of the connected loads. The SCHW Pumps shall be arranged in parallel together and connected via a common piping header; one for LTCHW and one for MTCHW." Remove 7.10.1 Remove 7.10.2 Table 6: Equipment Selection Criteria Chiller(s): Add starter type Distribution Pump(s): Remove "or vertical inline" Cooling Tower(s): Cooling Tower shall consist of the following materials of construction: Add items 4,5,6,7,8 Provide the Cooling Tower with the following options/accessories: Remove "or with installation of fire sprinkler"from "PVC fill and drift eliminators shall be Factory Mutual (FM) approved materials or with installation of fire sprinkler."	
	8.0	Table 7 – Pipe Sizing Criteria for CHW and CW Piping Update "CHW Pipe Size", "Maximum Velocity" & "Maximum pressure drop" Remove Table 8 – Pipe Sizing Criteria for Condenser Water Piping Remove 8.4.1 De-coupler Piping for Primary/Secondary type CHW	

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		Systems	
	9.0	9.1 MT/LT Switchover Chiller Change to "Consider providing one MT/LT Switchover Chiller" from "A minimum of one MT/LT Switchover Chiller shall be provided" 9.2 MT/LT Backup Heat Exchanger Change to "A minimum one MT/LT Backup Heat Exchanger shall be provided in case of low loading in early stage or during winter season." from "In areas with cold weather climate conditions, one MT/LT Backup Heat Exchanger shall be provided."	
	10.0	10.3 Heat Recovery Chillers Change "35-38°C, option to 40-43°C" from "40-43°C" 10.5 Refrigerant Selection Add "GWP < 15" Remove "R-513" Change to "R-514A" from "R-514" Add 10.6 VFD chillers	
	13.0	Table 9: Accessories to be installed Air Vents: Remove "or Manual Air Vents" from "Install Auto or Manual Air Vents at all high point locations of CHW and CW piping." Remove "and downstream" from "Installation of auto air vents require an isolation valve upstream and downstream of the vent for service and manual operation" Remove "between the vent and isolation valve" from "Install a	

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		section of clear PVC piping on the downstream side of the vent, between the vent and isolation valve for visual inspection."	
		Valves: Change to "It does not recommend installing balancing valves to consumer with modulating control valve." From "All balancing valves shall be provided with position indicators and maximum adjustable stops (memory stops)."	
	14.0	Add 14.7	
	16.0	Table 11: control valve Fail positions Add bypass line control valve item Remove 16.3.1	
	19.0	19.1.4 Heat Exchangers Remove"(0 to 50/60 Hz)" from "Verify Heat Exchanger Control Valve Function through the operating range (0 to 50/60 Hz)"	

Appendix 1: Template Form for Site Specific Deviation List

Site: _____

No	Specification Section and Item	Deviation Proposal	Scope Impact (Benefit, Cost, Schedule)	Remark	Approval: 1. Site 2. GFTT